Rank-27 Vivify/Sweep Scheduling Repair

Tactix engineering handoff

22 August 2026

Executive outcome

The suspicious Vivify variable-range policy was not present in Kissat and has been removed. The follow-on comparison now also closes the residual Sweep scheduling mismatch that remained after the semantic-dose and arena-lifecycle repair. Tactix used a strict-overrun predicate inside its Kitten-mode sub-solver where Kissat stops at equality. That one tick moved seed 9080 from the opening Sweep session into the resumed session, reset the deterministic Kitten phase stream at a different point, and made later equivalence and representative orders look independently wrong.

The production repair is general: Kitten-mode decision and Sweep outer refinement polls are inclusive, exactly where Kissat's source uses `>=`. The enclosing Sweep seed loop remains strict `>`, also as in Kissat, and BVE's older entry-counter budget keeps its existing strict contract. No formula-size range, benchmark identity, literal, rank, deadline, or tuned constant enters solver behavior.

The repaired candidate matches the retained Kissat trace for all 595 opening seeds, all 2,303 resumed seeds, and all 343 equivalence identities in discovery order. Release passes 72/72. The required matched 28B+5C guard passes 66/66, all models and proofs validate, and residue is zero. Immutable rank 27 still returns UNKNOWN at 1,200.004 seconds, so rank 27 remains open.

Why one comparator changed the whole schedule

The first candidate-only retained-tail trace, job `1637760`, showed that the opening Sweep sessions were identical for 594 seeds. Retained Kissat then visited external seed 9080; Tactix stopped and retained it. At conflict 3069, Tactix began with 9080 and visited 2,306 seeds, versus Kissat's 2,303 beginning at 9132.

Tactix ended the opening call at 1,000,001 embedded ticks. The relevant reference source has two different boundary rules:

$$\begin{array}{lll} \text{Kitten decision / Sweep refinement:} & \text{ticks} \geq \text{limit} & \Rightarrow \quad \text{stop,} \\ \text{enclosing Sweep seed loop:} & \text{ticks} > \text{limit} & \Rightarrow \quad \text{stop.} \end{array}$$

Tactix had used the second predicate inside Kitten as well. At equality it made one additional decision, propagated to 1,000,001, and therefore correctly failed the enclosing test before seed 9080. Kissat stops embedded search at exactly 1,000,000, sees that the enclosing strict-overrun test is still false, and visits 9080 in the same session.

Session placement is semantic, not accounting trivia. One Sweep invocation owns one embedded Kitten generator; clearing an environment advances it, while starting a new invocation resets it. Moving 9080 across that boundary shifted the phase/model stream for every later environment. The previously observed four-way equivalence order and representative rescheduling differences were therefore downstream consequences, not separate stack-order defects.

Implementation and focused evidence

The source change is deliberately small.

- `SubSolver::solve` uses `spent >= budget` only for `SubTickMode::Kitten`. The entry meter retains `>`.
- Sweep polls `>=` before and after whole backbone/partition flip passes, matching `kitten_ticks_limit_hit`.
- Inner flip loops and the enclosing seed loop retain their source-backed strict-overrun checks.

A focused unit test imports three variables with one ternary clause, reaches the Kitten budget exactly after the first decision propagation, and requires Budget before a second decision. The pre-repair comparator returned SAT after doing the extra work. No soundness failure or crash occurred, so the plan's Claude repair-delegation trigger did not fire.

Candidate-only job `1637761` gives the exact retained comparison:

Boundary	Tactix	Kissat artifact	Comparison
Opening seeds	595	595	exact identity/order
Resumed seeds	2,303	2,303	exact identity/order
Equivalences	343	343	exact identity/order
Opening final seed	9080	9080	same session
Resumed final seed	347737	347737	same ceiling tail

The repaired opening call finishes at 1,000,131 ticks: it is allowed to start 9080 at equality, spends 131 ticks, and then stops at the enclosing strict cutoff. The second call spends 19,189,801 ticks and preserves the complete retained seed/equivalence sequence. These are the intended layered predicates, not a new unexplained overrun.

Full correctness and performance guard

Matched campaign `benchmarks/runs/20260822-0200-BC-sweep-inclusive-limit/`, job `1637762`, ran the accepted pin and candidate on all 28 Tier-B fixtures plus the fixed five mechanism-relevant Tier-C fixtures under 16 GiB and 1,200 seconds.

Arm	valid	SAT / UNSAT	aggregate	peak RSS
Accepted pin	33/33	17 / 16	2202.153 s	3.234 GiB
Candidate	33/33	17 / 16	2868.252 s	3.036 GiB

All 34 SAT models and 32 UNSAT proofs verify. There are no invalid, missing, malformed, cleanup-bad, or residual-certificate tasks. The candidate ratio is 1.302476, improved from the preceding lifecycle candidate's 1.326436 but still not an accepted performance pin. The decisive long guard `T105.2.0` remains proof-verified UNSAT and improves from 1,043.578 to 1,007.801 seconds; its 1,267,849,928-byte proof was checked and deleted. Candidate `mp1-Nb5T15` remains checked SAT at 713.803 seconds.

Immutable rank-27 witness

Only after the complete guard passed was job `1637828` submitted. It returned UNKNOWN at 1,200.004 seconds. The runner recorded a 303,038,464-byte partial proof, SHA-256 `0d49d511...7426d1`, then deleted it and all derived scratch. Certificate residue is zero. Rank 27 is not added to the nine closed ranks.

This is a limitation, not a reason to revert the repair. The change closes a source-proven scheduler mismatch, restores exact retained behavior through the localized boundary, loses no guarded solve, and improves the guard aggregate. It simply does not by itself move this hard witness across the fixed timeout.

Discipline and next action

The immutable input SHA-256 is `5435b1b2...bb4e9`; the accepted binary is `b6c991e0...fb6f`; the candidate is `09a77159...17ae3`. The branch and `origin/main` were synchronized before work, surviving unrelated jobs and worktrees were left untouched, and no frozen reference solver was rerun. Diagnostics used only retained Kissat evidence and candidate Tactix jobs.

The next rank-27 localization begins after the now-exact first two Sweep sessions. It must retain the closed Vivify scope/deduction/interleaving, TransitiveReduce order, Sweep semantic/representation lifecycle, and inclusive Kitten limit. The shifted-seed and representative-order investigations must not be repeated. The next source change again owes the full 28B+fixed-C guard before another Tier-D witness.

The durable technical report is `benchmarks/reports/2026-08-22-tierd-rank27-sweep-inclusive-limit.md`. The resume surface is `PROGRESS.md`; durable mechanism decisions are in `docs/implementation-notes.md` and `docs/scheduling-parity-ledger.md`.